

1.3 Classwork: Arithmetic Series

1. The sum of the first n terms of an arithmetic sequence is

$$S_n = \frac{n}{2}(2u_1 + (n-1)d) \quad \text{or} \quad S_n = \frac{n}{2}(u_1 + u_n).$$

Consider the arithmetic series

$$3 + 7 + 11 + 15 + \dots$$

- (a) Find S_5 , the sum of the first five terms, by writing out the five terms and adding them.
- (b) Use the first formula above to find S_5 . Show the values you substitute.
- (c) Find S_{20} , the sum of the first twenty terms.

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